

HAMIA KHALIL

Machine Learning & Data Engineer | Full Stack Web Developer

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PROFESSIONAL SUMMARY

Motivated Computer Science Student with a **3.92** CGPA, specializing in Machine Learning, Data Engineering, and AI. Proven ability to design and deliver web applications and data driven solutions using Python, React, Django, dbt and SQL. Published IEEE author with hands-on project experience and a strong academic record. Eager to contribute technical expertise and problem solving skills to a forward thinking organization.

EDUCATION

University of Central Punjab (UCP), Lahore

Bachelor of Science in Computer Science (BSCS)

2023 – Present

CGPA: 3.92 / 4.0

TECHNICAL SKILLS

Languages: C++, SQL, Python, JavaScript, HTML, CSS, ReactJS, NodeJS, Tailwind CSS

AI / ML: Data Science, Machine Learning, Deep Learning, Data Analysis, Ensemble Learning, Feature Engineering, Data preprocessing

Data engineering: MySQL, PostgreSQL, dbt, KNIME, Power BI, Tableau, ETL, Data Warehousing, Data Modeling, Apache Airflow, Docker

Tools: Microsoft Visual Studio, SQL Server, Ubuntu, Visual Studio Code, Google Collab, Git, Figma, MIT, AWS (cloud), KNIME, Power BI, Tableau, PostgreSQL, Supabase, dbt

Soft Skills: Teamwork, Leadership, Collaboration, Communication Skills, Strong Numerical and Analytical Skills

Web dev: HTML5, CSS3, JavaScript, React.js, Next.js, Tailwind CSS, Bootstrap, Django, Rest APIs

PROJECTS

AI-Based Plant Disease Detection System (Final Year Project)

- Developing a deep learning model to detect plant diseases and nutrient deficiencies from images
- Applying ensemble techniques and image processing for high-accuracy predictions
- Including Full stack Web application in DJANGO backend and PostgreSQL

Machine Learning Multi-Class Classification System

- Built a Voting Classifier combining Logistic Regression, Random Forest, and SVM in Python
- Improved model prediction accuracy through ensemble learning techniques

Python Project for Data Engineering

- Developed a Python-based ETL pipeline to extract, clean, transform, and process global bank market-capitalization data using Pandas and web scraping
- Loaded transformed data into CSV and SQLite, performed SQL queries, applied currency conversions, and implemented ETL process logging.

E-Commerce ETL Pipeline Orchestration

- Built an end-to-end ETL pipeline using Python and Pandas to clean, validate, transform and load E-Commerce data into PostgreSQL.
- Orchestrated the pipeline with **Apache Airflow** and **Docker**, implementing task monitoring, logging and automatic retries for reliable workflow execution.

E-Commerce Data Engineering Pipeline

- Built an end-to-end E-Commerce data pipeline using Python and Pandas to clean, validate and transform multi-source order, customer and product data before loading it into PostgreSQL.
- Developed **dbt** staging, intermediate and mart models with **source()** and **ref()**, implementing data quality tests, documentation and analytics-ready customer revenue models.

Retail Sales Analysis using PostgreSQL

- Developed a SQL-based retail sales analysis project using PostgreSQL and a real-world dataset of 1,000 sales records.
- Performed business analytics using SQL queries, aggregations, CASE statements, date functions, and reporting to generate actionable sales and customer insights.

Task Management & Asynchronous Processing System

- Developed a Python Flask backend with PostgreSQL for task management, authentication, and RESTful API operations.
- Integrated Redis & Celery with asynchronous task processing and message queues to handle background jobs efficiently.

Sales & Business Intelligence Dashboard on POWER BI

- Built an interactive Power BI dashboard using cleaned sales data, implementing **data modeling, Power Query transformations, DAX measures, KPIs, and drill down analysis.**
- Analyzed revenue, profit, sales trends, product performance, and regional insights through **interactive dashboards and business focused visualizations**

E-Commerce Business Intelligence Dashboard on POWER BI

- Designed an end to end Power BI dashboard analyzing **orders, revenue, profit, products, customers, and delivery performance.**
- Used **Power Query for ETL, DAX for advanced calculations, and a star schema** to deliver interactive business insights.

ETL Pipeline using Python and Pandas

- Built an **E-Commerce ETL Pipeline** using Python and Pandas to extract, clean, transform, validate, and process CSV/JSON data.
- Implemented data quality checks, duplicate handling, data type conversion, joins, and sales aggregations to generate analytics-ready datasets.

SQL Analytics & Database

- Developed hands-on SQL expertise using PostgreSQL by solving real-world business queries involving data transformations.
- Building an interview-focused SQL portfolio with daily practice, documented solutions, and real-world database scenarios.

Snooker Club Management System

- Developed a full stack System using Next.js/React, TypeScript, Tailwind CSS for the frontend and Supabase for backend services, authentication, and API integration.
- Designed and managed the PostgreSQL database with Supabase, implementing Table Management, POS/Billing, Canteen Inventory, Staff Roles, Sales Tracking, and Reporting.

Full-Stack Food Delivery Web Application

- Developed a fully responsive web app using React and Tailwind CSS with dynamic UI components

WanderWise – Tourism Web Application

- Built a responsive tourism platform with React and Tailwind CSS, optimized for user experience

Retro Vision – AR-Based Historical Visualization App

- Designed an interactive AR application to visualize historical events, enhancing educational engagement on Figma

AWS-Based Cloud Infrastructure

- Designed and deployed cloud infrastructure on Amazon Web Services (AWS) using EC2, S3, IAM, VPC, RDS, CloudWatch, Elastic Load Balancer (ELB), and Auto Scaling. Configured secure networking, storage, identity and access management, monitoring, and scalable application deployment following cloud computing best practices.

SkillBridge AI – AI-Powered Employability Prediction & Career Recommendation Platform

- Developed an end-to-end AI-powered employability prediction and career recommendation system using Python, Scikit-learn, XGBoost, feature engineering, exploratory data analysis (EDA), and ensemble machine learning techniques.

Library Management System

- Developed using Object Oriented Programming principles, implementing encapsulation, inheritance, polymorphism, and abstraction to manage books, members, and borrowing operations.

Data Structures and Algorithms Project

- Designed and implemented Red-Black Tree and Trie data structures, demonstrating proficiency in self-balancing trees, prefix searching, and algorithm optimization

Thalassemia Patient Management System

- Designed and implemented a relational database using SQL, ERD, relational schema design, and normalization. Developed SQL queries, joins, views, and constraints to efficiently manage patients, doctors, appointments, and medical records.

Agile based Mobile Application Development

- Developed a mobile application using MIT App Inventor by following Agile methodology. Performed requirements analysis, designed UML artifacts including Use Case, Sequence, and Class Diagrams, created Data Flow Diagrams (DFDs), participated in software testing, and produced technical documentation throughout the Software Development Life Cycle (SDLC).

University Network Infrastructure

- Developed a scalable campus network in Cisco Packet Tracer, configuring LANs, VLAN segmentation, routing protocols, DHCP, switching, and network devices to provide secure and reliable connectivity across university departments.

FlipMart (E-Commerce Clone Website)

- Developed a responsive e-commerce website inspired by Flipkart, featuring product listings, category navigation, search functionality, and a user friendly shopping interface.

Voyage (Travel Website)

- Designed and developed a responsive travel website showcasing destinations, tour packages, booking information, and an interactive user experience

Dream Agency Website

- Built a modern business website for a digital agency, featuring service portfolios, project showcases, team profiles, contact forms, and responsive UI design.

PUBLICATION

MORL-Based Green SLO Framework for Dynamic Carbon, Latency and Energy-Aware Optimization in Cloud-Edge Systems – *IEEE, 2025*

- Proposed a multi-objective reinforcement learning framework optimizing carbon footprint, latency, and energy efficiency in cloud-edge environments (<https://ieeexplore.ieee.org/document/11272930>)

JOB EXPERIENCE

Data Engineering Mentor At Zee Outsourcing Solutions Pvt. Ltd. June 2026-Present

- Mentor aspiring data professionals in Data Engineering concepts, including SQL, ETL processes, data pipelines, tools POWER BI & knime and database management.
- Guide interns and students through hands-on learning activities, real-world examples, and technical problem-solving.
- Support learners in understanding database concepts, data architecture, and automated data workflows.

Data Visualization Virtual Experience Program | Forage August 2026

- Completed a virtual job simulation focused on data visualization and business analytics.
- Used Tableau and Excel to analyze business data, create calculated fields, and classify equality scores.
- Applied data visualization techniques to generate business insights and support data-driven decision making.
- Strengthened practical skills in Tableau, Excel, and data analysis through realistic business scenarios.

Data Analytics Job Simulation – Quantum August 2026

- Analyzed datasets to identify trends, patterns, and actionable business insights.
- Applied data analytics and problem-solving techniques to a real-world business scenario.
- Developed data-driven recommendations and communicated findings to support business decisions.

Data Visualisation: Empowering Business with Effective Insights – TATA August 2026

- Acted as a virtual consultant to CFO and CMO executives by transforming complex datasets into actionable business dashboards
- Cleaned and analyzed large datasets to build targeted visualizations that directly answered critical financial and marketing business questions

BCG Data Science Job Simulation – Forage August 28, 2026

- Used Python (Pandas, NumPy) to analyze client data, build data visualizations, and deliver a concise executive summary with actionable business insights
- Engineered and optimized a Random Forest model that achieved a 50% recall rate in predicting customer churn

VOLUNTEER EXPERIENCE

Volunteer – PSRD (Pakistan Society for Rehabilitation of the Disabled) UCP, Lahore – 2024

- Facilitated interactive educational workshops with differently abled children
- Strengthened communication and teamwork skills through consistent community engagement

CERTIFICATIONS & ACHIEVEMENTS

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| <ul style="list-style-type: none">• Microsoft UX Professional Design Certificate (2025)• SQL Foundations – Microsoft• Data Manipulation and Transactions in SQL Server – Microsoft• Python Project for Data Engineering – IBM• Introduction to Generative AI – Google• AI Fluency: Framework & Foundations – Anthropic• IEEE Published Author (2025)• Web Development Competition - 1st Place• Data engineering Course Certificate• Gold Medalist – Annual Sports Gala (2026)• Data Visualization – TATA | <ul style="list-style-type: none">• Organizer – Think2Code IEEE Event• Participant – AI/ML Codex 2.0 (2026)• Participation in Debate Competition - 2024• Participation in Code Rush - 2023, 2024• Participation in Sports - 2024• Won Quiz Competition - 2023• Participation in Society of the Year - 2023, 2024• Data Science Job Simulation – BCGX• Introduction to Data Engineering-IBM |
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